

$$= 0.75 \text{ J}$$

5 **B** As object is floating, Weight of object = Upthrust

$$\rho_b V_b g = \rho_l V_{\text{submerged}} g$$

$$\rho_l = \frac{\rho_b V_b}{V_{\text{submerged}}} = \frac{(7.8)(l^3)}{\left(l^2 \times \frac{7}{8}l\right)} = 8.9 \text{ g cm}^{-3}$$

6 **D** Power input =

$$(400)(9.81)(1000) \text{ (1)}$$